NYU TANDON SCHOOL OF ENGINEERING

NYU TANDON CAREER HUB · FALL 2026

Data Science Bootcamp

Week 2 · Numpy & Pandas

Instructor: **Angelin Ann Jacob**

Section 1 · Tue Oct 6

Section 2 · Thu Oct 8

Useful Links

Zoom
All sessions are virtual. Use ONLY **your NYU SSO Login to be considered for attendance**

Brightspace
Assignments, submissions and the discussion board, monitored Monday to Thursday.

GitHub
Slides and Colab notebooks for every session.

Email
datasciencebootcamp@nyu.edu for questions, extensions and attendance approvals.

1:1 coaching
Optional 30-minute technical interview prep. Book it through Handshake.

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Numpy & Pandas

Get to the libraries used in your day to day tasks as a Data Specialist!

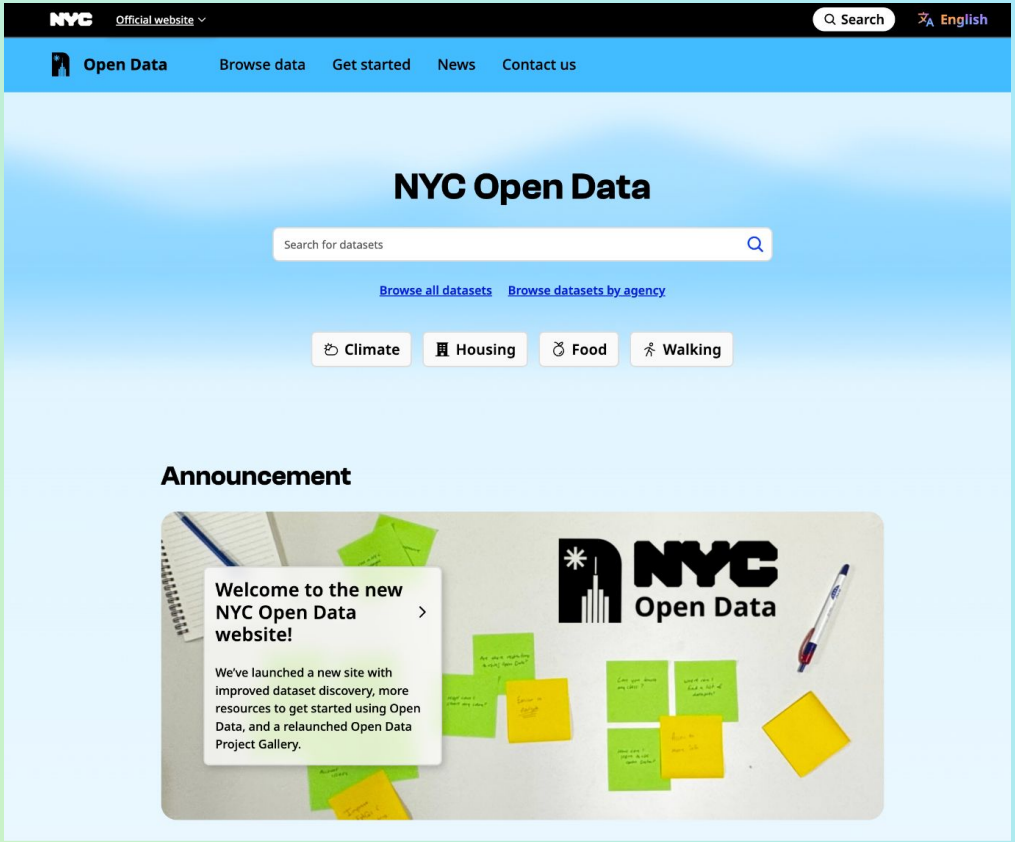
Data Science Bootcamp · Fall 2026

NYC Open Data

https://www.nyc.gov/opendata

Dataset name:
Brooklyn Bridge Automated Pedestrian Counts Demonstration Project (Historic)

Data Provided By:
Department of Transportation (DOT)
Visit: NYC Open Data - City of New York - Brooklyn Bridge Automation



HOW TO READ TODAY'S SESSION

Two parts for every code block

THE STORY

The question about the bridge

What we want to know, in plain English.

When do people cross? Which direction dominates? Does rain matter?

THE TOOL

The library and function

How we get the answer, in code.

```
np.reshape()    pd.read_csv()
df.groupby()    df.sort_values()
```

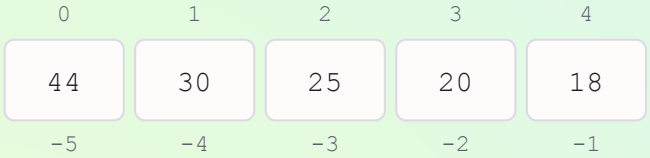
EXAMPLE

Story: Which hours of the day are busiest?

```
bridge.groupby("hour")
      .pedestrians.mean()
```

PART 1 · NUMPY · INDEXING & SLICING

Pick out exactly what you need



`pedestrians[:5]`

Five real counts, two ways to number them: from the front, or from the back.

Spot-check random hours with **`np.random.randint`**.

THE STORY

Which hour was recorded first? Which last?

THE TOOL

```
pedestrians[0]
pedestrians[-1]
direction[0, 0]
```

THE STORY

Pull out one full day, or the same hour every day.

THE TOOL

```
a[:24]      # one day
a[::24]     # every 24th
a[::-1]     # newest first
```

THE STORY

Edit a slice. Does the original change?

THE TOOL

```
view = a[:5]
view[0] = -1   # a too?!
a[:5].copy()
```

PART 1 · NUMPY · RESHAPING & AXES

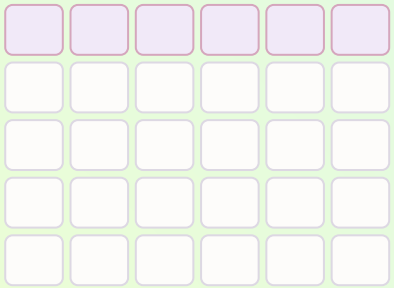
Turn hours into days



7,296 hourly counts in one row

`reshape((304, 24))`

24 hours →



304 days ↓

axis=0

collapse the days: one typical day

axis=1

collapse the hours: one total per day

THE STORY

Which hours are busiest on a typical day?

One row per day, one column per hour.

Encoding columns and rows with positional info!

THE TOOL

```
grid = ped.reshape((304, 24))
grid.mean(axis=0)
grid.sum(axis=1)
typical_day[np.newaxis, :]
np.where(cond, "busy", "quieter")
```

PART 2 · PANDAS · THE TABLE

Meet the table

THE STORY	THE TOOL
What did we just load? How big is it?	`bridge.head()` `bridge.tail()` `bridge.shape` `bridge.info()`
Typical values. Is the mean far from the median?	`bridge.describe()` `bridge.dtypes`
How many people in total? How common is each weather?	`bridge["Pedestrians"].sum()` `bridge["weather_summary"].value_counts()`
Busiest hours? Rainy versus dry? The single busiest hour?	`bridge.sort_values(ascending=False)` `bridge[bridge["precipitation"] > 0]` `bridge.loc[bridge["Pedestrians"].idxmax()]`

PART 2 · PANDAS · GROUPING

Group, compare, summarize

Every hourly row

one row = one hour

→

```
analysis.groupby("hour")
```

split into 24 piles, one per hour

→

```
.pedestrians.mean()
```

one number per pile: a Series

THE STORY

Tidy names, then pull out the hour.

THE TOOL

```
df.rename(columns={...})
df.pedestrians
df["time"].dt.hour
```

THE STORY

Mean, min and max for every weather label.

THE TOOL

```
.groupby("weather_summary")
.agg([...])
.describe()
```

THE STORY

Readable labels, and tables we build ourselves.

THE TOOL

```
.apply(lambda q: ...)
pd.Series(...)
pd.DataFrame({...})
```

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Before next week

Take-home

The exercises at the end of today's notebook. Submit on Brightspace within one week, by 11:59 PM.

Feedback form

Fill it out after today's session. It counts toward your credential.

Week 3: EDA and Data Exploration

Work with datasets and explore trends and data stories.